Persona Name	Module Name	Sub Topic	SME Hours	Hands-on Hours	E-Learning Hours	Total Hrs
Cloud Data Enginee	**1.Introduction to Data Warehousing**	Basics : Overview of Data Warehousing; Role of Data Engineer vs BI Developer vs Analyst; OLTP vs OLAP Systems; Data Warehouse vs Data Lake vs Lakehouse; Use Cases of Data Warehousing	2	0	1	3
	2.Data Warehouse Architecture	Architecture : 3-Tier Architecture (Source, Staging, Warehouse, BI Layer); Data Ingestion Layer; Staging Layer Design; Enterprise Data Warehouse (EDW); Data Marts (Dependent vs Independent); Modern Architectures (Lambda, Kappa, Lakehouse)	1	1	2	3
	3.Data Modeling	Fundamentals : Types of Data Models (Conceptual, Logical, Physical); ER Modeling Basics; Entities and Attributes; Relationships (1-1, 1-M, M-M); Normalization (1NF, 2NF, 3NF); Denormalization Techniques, Data Cleansing and manipulations	2	2	1	5
	4.Data Injection & Processing	ETL/ELT concepts : Why & How ETL/ELT drive Data Pipelines construct, Schema on Write vs. Schema on Read, Batch vs Streaming, Differences and common tools	2	2	1	5
	5.Business Intelligence	Introduction to Analytics & Concepts : Overview of Data visualization, Dashboards, Scorecards, Reports (Dynamic & Static), Self-service reporting and leading tools	1	1	1	3
	6.Dimensional Modeling	Types of Schemas : Star Schema; Snowflake Schema; Galaxy Schema	1	1	1	3
	7.Dimensional Modeling	Fact Tables : Transactional Fact Tables; Snapshot Fact Tables; Accumulating Snapshot Tables; Measures (Additive, Semi-additive, Non-additive)	2	2	1	5
	8.Dimensional Modeling	Dimension Tables : Conformed Dimensions; Role-playing Dimensions; Junk Dimensions; Degenerate Dimensions	2	2	1	5
	9.Keys and Relationships	Keys : Primary Key; Foreign Key; Surrogate Key vs Natural Key; Composite Keys	1	1	1	3
	10.Keys and Relationships	Integrity : Referential Integrity; Constraints; Handling Missing Keys	1	1	1	3
	11.Slowly Changing Dimensions (SCD)	SCD Types : SCD Type 0; SCD Type 1; SCD Type 2; SCD Type 3	1	2	1	4

	12.Slowly Changing Dimensions (SCD)	Concepts & use : Surrogate Key Generation; Effective Start and End Dates; Current Flag; Versioning	2	2	1	5
	13.Data Warehouse Design	Design Principles : Grain Definition; Fact-Dimension Relationship Design; Hierarchies (Time, Geography); Data Sparsity Handling; Aggregation Tables	2	2	1	5
	14.Data Warehouse Optimization	Performance management : Partitioning (Range, Hash); Indexing (B-tree, Bitmap); Clustering; Materialized Views; Query Performance Tuning Basics	2	2	1	5
	15.Data Quality & Handling	Concepts & Practices : Understanding data profiling, cleaning, validation, Standardization, Lineage and establishing DQ metrics Structured, Semi-Structured and Unstructured data examples and handling	2	2	2	6
	16.Metadata and Governance	Governance overview : Technical Metadata; Business Metadata; Data Lineage; Data Catalog Basics; Data Governance Principles; Data Security Basics	1	1	1	3
	17.SQL Fundamentals	Basics : SELECT Statement; WHERE Clause; ORDER BY; DISTINCT; LIMIT	1	2	1	4
	18.NoSQL	Fundamentals : SQL vs NoSQL, Horizontal Scalability, Dynamic Schema, CRUD Operations	1	1	1	3
	19.SQL Joins	Joins : INNER JOIN; LEFT JOIN; RIGHT JOIN; FULL JOIN; Self Join	2	2	1	5
	20.SQL Aggregations	Aggregation : GROUP BY; HAVING Clause; Aggregate Functions (COUNT, SUM, AVG, MIN, MAX)	2	2	1	5
	21.SQL Advanced	Subqueries & CTE : Subqueries (Correlated and Non-correlated); Common Table Expressions (CTE)	2	3	1	6
	22.SQL Advanced	Window Functions : ROW_NUMBER; RANK; DENSE_RANK; LEAD; LAG	2	4	1	7
	23.SQL Data Transformation	Transformations : Joins for Data Transformation; CASE Statements; Data Cleaning using SQL; Handling NULL Values	2	3	1	6
	24.SQL for Semi-Structured Data	JSON Handling : Working with JSON Data; Parsing JSON; Querying Nested Data	1	2	1	4
	25.SQL Performance Optimization	Optimization : Indexing Concepts; Query Optimization Basics; Execution Plan (Intro); Partition Pruning	2	2	1	5

	26.PL/SQL / Procedural SQL	Programming : Stored Procedures; Functions; Packages (Intro)	2	3	1	6
	27.PL/SQL / Procedural SQL	Control Flow : Cursors; Loops; Conditional Statements	1	2	1	4
	28.PL/SQL / Procedural SQL	Error Handling : Exception Handling; Logging Errors	1	2	1	4
	29.PL/SQL / Procedural SQL	Automation : Triggers; Using Procedures in ETL Pipelines	1	2	1	4
	30.SQL for Data Engineering	Automation : Writing SQL for ETL Pipelines; Data Validation Queries; Data Reconciliation Queries; Incremental Data Processing	2	6	1	9
	31.Introduction to Python	Basics : Overview of Python and Its Applications; Installing Python and Setting Up Environment (IDLE, VSCode); Writing and Running Python Scripts; Understanding Python Syntax and Indentation; REPL vs Script Execution	1	1	1	3
	32.Python Essentials	Core Concepts : Variables and Data Types; Input and Output Operations; Operators (Arithmetic, Comparison, Logical, Assignment); Comments and Code Readability; Type Casting; Difference between "is" vs "=="	2	2	1	5
	33.Control Flow	Decision Making & Loops : Conditional Statements (if, elif, else); Loops (for, while); Loop Control Statements (break, continue, pass); Match-case (Structural Pattern Matching); Nested Loops and Conditions; enumerate() and zip()	2	2	1	5
	34.Functions and Modules	Functions : Defining and Calling Functions; Function Arguments and Return Values; Default and Keyword Arguments; *args and **kwargs; Returning Multiple Values	2	2	1	5
	35.Functions and Modules	Modules & Scope : Importing and Using Modules (Built-in and Custom); Module Structure and Imports; Variable Scope (Local vs Global); Docstrings	1	1	1	3
	36.Data Structures in Python	Lists & Strings : Lists; Creating and Manipulating Lists; List Comprehensions; Working with Strings; Creating Strings; String Concatenation; Multi-line Strings; String Formatting; Accessing Characters; String Slicing; String Methods	2	2	1	5

	37.Data Structures in Python	Advanced Data Structures : Tuples and Their Usage; Dictionaries (Key-Value Manipulations); Sets and Unique Element Operations; Mutable vs Immutable Types; Deep Copy vs Shallow Copy; Dictionary and Set Comprehensions	2	2	1	5
	38.File Handling	File Operations : Reading and Writing Files; Working with Text and CSV Files; Exception Handling in File Operations; Reading Large Files (Line-by-Line); Working with JSON Files; File Paths using os and pathlib	2	2	1	5
	39.Object-Oriented Programming (OOPS)	OOP Concepts : Introduction to OOP Concepts; Defining Classes and Objects; Constructors and Methods; Inheritance and Polymorphism	2	2	1	5
	40.Python Libraries and Tools	Data Libraries : NumPy Basics and Array Operations; Pandas DataFrames and Series; Reading and Writing CSV and JSON; Filtering, GroupBy and Joins; Handling Missing Values	3	4	1	8
	41.Python Libraries and Tools	Utility Libraries : Matplotlib Basic Plotting; Datetime Module; Requests (API Calls); OS Module	1	1	1	3
	42.Error Handling and Debugging	Exception Handling : Types of Errors in Python; Try, Except, Finally Blocks; Custom Exceptions; When to Catch vs Not Catch Exceptions	2	2	1	5
	43.Error Handling and Debugging	Debugging : Debugging Techniques and Tools	1	2	1	4
	44.Unit Testing with Python	Testing : Organizing Tests with TestCase Class; Creating Test Cases; Running unittest Tests; Exploring Assert Methods	1	2	1	4
	45.Logging with Python	Logging : Python Logging Module; Configuring Logging System; Logging Levels; Logging vs Print; Common Logging Pitfalls and Solutions	1	2	1	4
	46.Advanced Topics	Functional & Advanced Concepts : Lambda Functions; map(), filter(), reduce(); Iterators and Generators; Decorators (basic); Working with Regular Expressions	2	4	1	7
	47.Advanced Topics	Data & Integration : Python JSON (Important); API Calls; Python API and Web Scraping (basic); Working with Python SQLite (basic)	2	2	1	5
	48.Advanced Topics	Environment & Packaging : Introduction to Virtual Environments; Package Management using pip	1	1	1	3

	49.GitHub Copilot – Utilization	Copilot Basics : GitHub Copilot Overview; Basic Copilot Usage; Understanding Copilot Suggestions; Copilot in IDEs; GitHub Copilot Language Support	1	1	1	3
	50.GitHub Copilot – Utilization	Advanced Usage : Generative Commit Messages; Copilot Chat; Copilot for CLI; Code Documentation; Code Optimization; Refactoring Strategies; Debugging with Copilot; NLP with Copilot	1	1	1	3
	51.GitHub Copilot – Utilization	Best Practices : Copilot Limitations and Strengths; Troubleshooting Issues; Copilot Best Practices; Usage in Existing/Legacy Code	1	1	1	3
	52.Introduction to Pyspark	Fundamentals & Setup : Introduction to PySpark; Spark Architecture (Driver, Executor, Cluster Manager); Spark Setup & Environment; Running PySpark Applications; Spark UI Basics	2	2	1	5
	53.Introduction to Pyspark	Core Processing : RDD (Basics & Operations); DataFrames (Creation, Transformations, Joins, Aggregations); Spark SQL (Queries, Views, Window Functions); File Handling (CSV, JSON, Parquet, Data Lake Integration)	6	8	2	16
	54.Gen AI	Fundamentals : Generative AI Fundamentals; NLP Basics	1	1	1	3
		Prompt Engineering : Prompt Engineering Techniques	2	3	1	6
		LLM Integration : Working with LLM APIs using Python	1	3	2	6
		Data Engineering Use Cases : GenAI for SQL Query Generation; Data Query Automation; Data Summarization and Insight Generation	1	3	2	6
		Advanced GenAI Concepts : Retrieval Augmented Generation (RAG) for Data Systems; GenAI with Pandas and DataFrames	1	3	2	6
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	55.GCP Data	Basics of GCP : Hierarchy and Resource Management (Organization, folders, projects, labels, etc.); Regions and Zones; Identity and Access Management; Cloud Console, Cloud Shell; Monitoring; Logging; Error Reporting; Managing virtual machines using Console and Shell (gcloud); Accessing Virtual Machines using SSH; Preemptible VMs; Security with User Managed Service Accounts	3	4	1	8
		Cloud Storage : Managing Buckets using Console and Shell (gsutil); Storage Classes; Transferring Data; Object Versioning; Managing Object Lifecycle; Managing Security	2	3	1	6

		Dataflow : Building Batch Data Pipelines using Apache Beam; Building Streaming Data Pipelines using Apache Beam; Beam and Dataflow SQL; Parametrizing Pipelines; Deploying Pipelines as Dataflow Jobs using Shell; Monitoring Dataflow Jobs	4	8	2	14
		BigQuery : Interacting with BigQuery using Console, Python, and Shell (bq); Loading Data into BigQuery; Working with Datasets; Working with Native Tables; Working with External Tables; Partitioning; Clustering; Views; Materialized Views; Using Cached Results; Running Parameterized Queries; Scheduling Queries; Querying External Data Sources; Managing Security	4	8	2	14
		Dataproc : Running Hadoop on Cloud Dataproc; Using GCS instead of HDFS; Managing Dataproc Clusters using Console and Shell; Submitting Apache Spark Jobs on Cloud Dataproc; Parametrizing Spark Jobs; Using Workflows to Manage Lifecycle of Clusters	3	6	1	10
		Cloud Composer : Fully Managed Apache Airflow Implementation; Apache Airflow: Programmatically Create, Schedule, and Monitor Data Workflows	2	4	1	7
	56.Azure Data	Basics of Azure : Resource Groups; Subscriptions; Regions; Identity and Access Management (Azure AD, RBAC); Azure Portal, Azure CLI; Monitoring (Azure Monitor); Logging (Log Analytics); Virtual Machines (creation and management); Accessing VMs using SSH; Managed Identities	2	3	1	6
		ADLS Gen2 : Creating Storage Accounts; Containers and File Systems; Storage Tiers; Uploading and Transferring Data; Access Control (ACLs, RBAC); Lifecycle Management	2	3	1	6
		Azure Data Factory : Pipelines; Activities and Data Flows; Linked Services; Datasets; Triggers and Scheduling; Monitoring Pipelines; Parameterization; Dynamic Content; Nested Pipelines; Mapping Data Flows	4	8	2	14
		Azure Databricks : Workspace and Clusters; Notebooks; Integration with ADLS	2	3	1	6
		Databricks Fundamentals : What is Databricks; Lakehouse Architecture; Workspace Walkthrough (Notebooks, Clusters, Workflows); Unity Catalog Basics	2	3	1	6

		Spark : Spark Architecture; DataFrames; Spark SQL; Transformations and Actions; Structured Streaming Basics	3	6	1	10
		Delta Lake : What is Delta Lake; Creating Delta Tables; Schema Versioning; Time Travel in Delta Lake; Data Integrity (ACID Transactions)	2	4	1	7
		Unity Catalog : What is Unity Catalog; Unified Governance Architecture; Modern Data Governance; Data Quality Management; Lineage, Auditing and Metadata Management	2	3	1	6
		Data Engineering with Databricks : Data Ingestion with Lakeflow Connect; Deploy Workloads with Lakeflow Jobs; Build Pipelines with Lakeflow Declarative Pipelines; Data Validation Techniques; Handling Missing and Invalid Data; Data Quality Checks in ADF and Databricks; Logging and Monitoring Pipelines; Alerting and SLA Management	3	6	1	10
		Gen AI for Data Engineering : Generating SQL Queries using LLMs; Data Summarization and Insight Generation; Automating Data Documentation; GenAI with Pandas and DataFrames; Building AI-powered Data Assistants; Integrating Azure OpenAI with Data Pipelines	2	3	1	6
	57.AWS Data	Basics of AWS : AWS Account Structure; Regions and Availability Zones; Identity and Access Management (IAM); IAM Roles and Policies; AWS Console and CLI; Monitoring (CloudWatch); Logging (CloudTrail); Error Monitoring; EC2 Virtual Machines; Accessing EC2 using SSH	2	3	1	6
		Amazon S3 : Buckets; Storage Classes; Uploading and Transferring Data; Versioning; Lifecycle Policies; Security and Access Control	2	3	1	6
		Data Warehousing (Redshift) : Cluster Setup; Loading Data (COPY command); Querying; Distribution Keys and Sort Keys; Performance Optimization; Redshift Spectrum (Querying S3); Security	3	5	1	9
		AWS Glue : Crawlers; Data Catalog; ETL Jobs; Transformations (PySpark); Job Scheduling; Monitoring Jobs; Schema Inference	4	8	2	14
		AWS Glue Studio : Visual ETL Interface; Drag-and-drop Pipelines; Job Debugging	1	2	1	4

		Data Lake Architecture : Building Data Lake on S3; Layered Architecture (Raw, Curated, Trusted); Data Lake Governance Basics	2	3	1	6
		Streaming (Kinesis) : Data Streams; Data Firehose; Real-time Data Processing; Integration with S3, Redshift	2	4	1	7
		Serverless Data Processing : AWS Lambda; Event-driven Processing; Integration with S3, Kinesis, APIs	2	3	1	6
		Workflow Orchestration : Step Functions; State Machines; Pipeline Automation	1	2	1	4
		Apache Spark on AWS : Spark on EMR (Elastic MapReduce); Cluster Management; Running Spark Jobs; Integration with S3	3	5	1	9
		Query Services : Amazon Athena; Querying Data in S3 using SQL; Partitioning; Performance Optimization	2	3	1	6
		Data Governance : AWS Lake Formation; Data Catalog Governance; Access Control; Data Security	1	2	1	4
		Data Migration : AWS DMS (Database Migration Service); Bulk Data Transfer; Continuous Replication	1	2	1	4
		API & Ingestion : API Gateway; Data ingestion from APIs; Integration with Lambda	1	2	1	4
		DevOps for Data Engineering : CI/CD Basics; Infrastructure as Code (CloudFormation/Terraform intro); Deployment of Data Pipelines	2	3	1	6
		Gen AI on AWS : Amazon Bedrock; LLM Integration; Prompt Engineering; AI-powered Data Pipelines; Generating SQL and Insights	2	3	1	6
	58.Capstone Project	End-to-End Capstone Project : Build an end-to-end pipeline using file/API ingestion, transformation, storage, SQL analytics, and one GenAI use case	2	24	4	30
